



Java OOPs: Inheritance and Overriding Assignment - 2

Problem Statement:

You are working on a Java program to model geometric shapes. Your task is to create a base class "Shape" with methods to calculate the area and perimeter of different shapes. Then, you need to create two derived classes, "Rectangle" and "Circle," which inherit from the Shape class. Each derived class should override the area and perimeter methods to provide their own implementations.

Once the classes are defined, you need to write a program to demonstrate polymorphism by using the base class reference to access the derived class methods.

Class: Shape

Tasks To Be Completed:

1. In the main program, create objects of the Rectangle and Circle classes.
2. Set the properties of each object.
3. Use the base class reference to access the calculateArea() and calculatePerimeter() methods of both the Rectangle and Circle objects.
4. Display the calculated area and perimeter for each shape.